Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 4.1 · Two secants

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 - 1$. Let A be the point of its curve with x-coordinate 1.

1. Calculate the slope of the secant joining A to the point of the curve with x-coordinate 2.
2. Repeat the calculation with the point of the curve with x-coordinate 1.1.
3. After studying the derivative at a point, determine $f'(1)$ and compare this value with the two rates calculated. Explain what changes when the second point gets closer to A.

Exercise 4.2 · A tangent

Consider the function $f(x) = 2x^2 + 1$, defined on $\mathbb{R}$. The aim is to determine the equation of its tangent at the point with x-coordinate -1, starting from the rate of change.

1. For $h \neq 0$, write then simplify $\dfrac{f(-1+h) - f(-1)}{h}$.
2. Deduce $f'(-1)$. Also calculate $f(-1)$.
3. Write the slope-intercept equation of the tangent and check that it passes through the point of contact.

Exercise 4.3 · Marginal cost

A company uses the model $C(q) = q^2 + 20q + 100$, where $q \geq 0$ is the quantity produced and $C(q)$ its cost in monetary units. The model extends the quantity to a real variable so as to study small changes in production.

1. Calculate $C(10)$ and $C(11)$, then the rate of change of C between 10 and 11. Interpret this rate in context.
2. From the rate between 10 and $10 + h$, determine $C'(10)$.
3. Compare the two results. Explain why the cost of one extra unit is not exactly equal to the derivative in this model.

Exercise 4.4 · Reading a tangent

This exercise has two independent situations. A tangent to a curve is a line: another point of this line does not necessarily belong to the curve.

1. On the figure in the section on the tangent, read the coordinates of two points of the green line. Deduce its slope and check the value stated in the legend.
2. In another situation, the tangent to the curve of a function f at the point $A(2,3)$ passes through $B(4,2)$. Determine $f(2)$ and $f'(2)$. Can we deduce $f(4)$?
3. Use this tangent to suggest an approximate value of $f(2.1)$. Make the "approximately equal" sign appear in the conclusion.

Exercise 4.5 · Checking an approximation

We wish to approximate 0.98^2 without first carrying out the multiplication. We use the tangent to the curve of the square function at the point with x-coordinate 1.

1. Write the equation of this tangent and deduce the required approximation.
2. Then calculate the exact value of 0.98^2. Define the error as "exact value minus approximate value": calculate this error and state its sign.
3. Explain the result using the identity $(1+h)^2 = 1 + 2h + h^2$.

Exercise 4.6 · Tangent and intersection

The function f is defined on $\mathbb{R}$ by $f(x) = x^3$. We accept that $f'(1) = 3$. A student claims that a tangent and a curve always have a single common point.

1. Determine the equation of the tangent to the curve of f at the point with x-coordinate 1.
2. Check by expanding that $x^3 - 3x + 2 = (x-1)^2(x+2)$.
3. Determine all the points of intersection of the curve and this tangent. Is the student's claim correct?

Exercise 4.7 · Numerical secants

To study the square function near 1, define, for $h \neq 0$,

$$T(h) = \frac{(1+h)^2 - 1}{h}.$$

We will use the six values 0.1, 0.01, 0.001 and their opposites.

1. Write a Python function that calculates $T(h)$, then produce a table of six results.
2. Conjecture the value of the derivative of the square function at 1.
3. Simplify $T(h)$ algebraically to justify the conjecture. Explain why trying numerical values is not, on its own, enough to prove the result.

Exercise 4.8 · Delayed review

The reciprocal function is defined by $f(x) = \dfrac{1}{x}$ for $x \neq 0$. We study it near 1.

1. State the forbidden values of h in the rate $\dfrac{f(1+h) - f(1)}{h}$, then simplify it.
2. Deduce $f\prime(1)$ and the equation of the tangent at 1.
3. Use this tangent to approximate $\dfrac{1}{1.02}$. Keep the distinction between exact value and approximate value.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 17 p. 127, all questions. Read two points on each tangent; write the quotient of the increments rather than estimating its steepness.

CQFD, exercise 19 p. 127, questions 1 and 2. Distinguish the y-coordinate of the point of contact from the slope of the tangent line.